
FlatPixel

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1 Introduction

FlatPixel is a tool for generating diffuse and normal spritesheets from 3d models. Generated sprites can be used for a variety of applications, however the intention of this project is to use them in 2d game engines, especially those with built-in 2d lighting solutions.

This document serves to explain the various features and options available within FlatPixel. In addition, tips and tricks for various common patterns are explained to better show the capabilities of the tool.

2 Settings Panel

The Settings Panel (located on the left side of the main UI) is the primary location where renders are defined. This panel has four tabs, labelled **Model**, **Camera**, **Render**, and **Export**.

2.1 Model Tab

The Model tab is responsible for loading models and selecting animations for render. Currently, the tool only supports models in the **glTF** and **glb** formats.

2.1.1 Animation Selector

If the imported model contains animation tracks, the list of animations will be provided below the model import button. Checking off an animation will mark it for render, and any render processes will render *all* animations into their spritesheet. If no animation is provided, rendering will default to the animation currently selected in the Preview Panel.

Models without animation tracks are also supported, but no animations will be listed and any renders will instead be a single still image.

2.2 Camera Tab

In order to produce renders of 3d models, a Camera is used. The Camera settings are used to adjust the parameters of the camera that exists within the 3d simulation/capture space.

2.2.1 Model Rotation

The **Model Rotation** slider defines the yaw rotation of the model in space. It is defined in degrees. The slider allows rotation between -180 and 180 degrees, which allows for any angle to be selected.

2.2.2 Camera Position

The **Camera Offset** and **Camera Tilt** options define the position of the camera within the scene. For orthographic cameras, the Z component of the camera offset effectively only impacts the clipping plane of the camera. However, perspective cameras will see a change in image based on its distance to the model.

The tilt of the camera is defined on the pitch axis.

2.2.3 Projection Modes

The camera defaults to an Orthographic projection mode, however a Perspective camera is also available. Depending on the projection selected, other options are also available.

For Orthographic cameras, the **Camera Size** parameter is available, which acts as a zoom. Higher size values lead to a larger area being captured.

For Perspective cameras, the **Field of View** parameter is available. This sets the FoV of the camera. Try different combinations of FoV and Z offset to get different camera effects.

2.2.4 Autocenter

Clicking the **Autocenter** button will attempt to center the camera on the model. This will only adjust the **Camera Offset**, but may behave strangely for some models.

2.3 Render Tab

The Render tab provides options pertaining to the actual rendering performed by the camera in the 3d scene.

2.3.1 Render Resolution

The render resolution of each frame within the spritesheet. Non-square resolutions are allowed and can be used to produce sprites that are on rectangular grids.

2.3.2 Render FPS

When an animation is rendered, a fixed number of sprites are created per "animation second". To achieve this, the animation is manually stepped by increments according to the frames per second. Each sprite is created with a time difference of $1 / \text{FPS}$ seconds in between.

Higher FPS values will lead to smoother animation, but larger spritesheets and longer render times.

2.3.3 Color Quantization

Enabling Color Quantization will force the generated image to only use colors from the provided palette. This effect is enabled only when the checkbox is set and the palette contains at least a single color. Colors can be added to the palette using the "Add Color" button. Clicking the "X" next to a color will remove it from the palette. To change a color, click on the color within the palette and use the color picker.

Palettes can be loaded from images. When an image is provided, every pixel will be read and the collection of distinct colors found will become the current palette. During this process, alpha values are ignored. For best results, use images in PNG format with a single pixel of each desired color.

Finally, palettes can be saved. This will produce a palette in the ideal format for loading, as described above.

2.3.4 Render Spritesheet

Clicking the Render Spritesheet button will produce a render given all current parameters. When rendering, animations from the Model Tab will be used, unless no animations are selected. When no animations are selected, the currently previewed animation is rendered instead. Objects with no animations will have a still image rendered.

The renderer will always run twice - once for diffuse and once for normals.

2.4 Export Tab

Rendered spritesheets are exported as PNG files to a location of the user's choosing. Optionally, normal maps can be exported alongside the diffuse textures (diffuse texture is always included). In addition, an info file containing metadata about the export, the spritesheet, and contained animations will be produced alongside the spritesheet(s).

2.4.1 Export Location

Selects the base export location for the diffuse texture file. Click the "... " button to open a file dialogue.

Normal maps will be exported to `<base-file>-normals.png`, and the info file will be exported to `<base-file>.json`.

2.4.2 Spritesheet Columns

Number of columns in the exported spritesheets. The number of rows is automatically calculated from this value and the number of total rendered frames.

2.4.3 Export Normal Maps

Checking this box will cause the normal maps to be exported alongside the diffuse spritesheet, at the path described above.

2.4.4 Export

Clicking this button will execute the Export. In addition, if any other settings have changed between the last render and the export, a render will automatically be run to ensure the exported spritesheets are in their most up-to-date state.

3 Preview Panel

The Preview Panel provides a view into what the camera can currently see, or what the most recent render looks like.

3.1 Preview

By default, the preview panel is open to the camera preview, which shows the current model with the current camera and render settings applied.

This is an accurate representation of what will be produced when rendering, with the exception of the "Frames per Second" option - the preview window will run at the native application FPS instead of the selected value.

3.1.1 Preview Animation

Picking an animation in this dropdown will cause it to play in the preview window. In addition, the animation can be scrubbed with the slider, or played and paused.

If no animation is selected in the Model tab when rendering, the animation currently selected here will be used instead.

3.1.2 Preview Normals

Toggles the preview window to show the normal maps instead of the diffuse texture.

3.2 Spritesheet

After rendering, this tab will be populated with the results of the most recent render. This can be used to inspect generated sprites before exporting.

3.2.1 Sprite Type

With this dropdown, both the diffuse and normal spritesheets can be inspected.

4 Saving Sessions

FlatPixel allows the current user session to be saved to disk as a `.flpx` file. This file contains all of the relevant information to recover and continue working on a given capture project.

Because `flpx` files retain *references* to the currently opened model on disk, they are not generally able to be sent between devices. However, should a model be missing during the load process, the user will be prompted to relocate the model (and in doing so can successfully load a `flpx` they have received from someone else, provided they also have the corresponding model file).

Session files can be opened and saved from the top toolbar of the application.